

can check using commands given below (also shown in Fig. 3).

```
info.DeviceInfo(1)
info.DeviceInfo.SupportedFormats
```

In Fig. 3, you can see that there are many formats that your camera can support. When you get the supported formats and device number, you can make changes in the source code as per your requirement.

Some important functions used in the program are:

data = getsnapshot(vid). This function immediately returns one

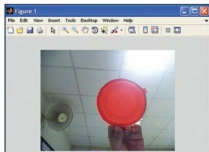


Fig. 4: Snapshot of the original image

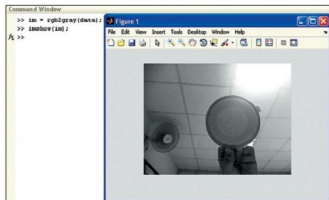


Fig. 5: RGB to greyscale

single-image frame from the video input object as shown in Fig. 4.

img = rgb2gray(data). It converts the true RGB-colour image to greyscale intensity image *img*. And *rgb2gray* converts RGB images to greyscale by eliminating the hue and saturation information, while retaining the luminance as shown in Fig. 5.

data(:, :, 1). It extracts all red-colour components from the original image. It does the same for green data(:, :, 2) and blue data(:, :, 3), as shown in Fig. 6.

diff_im = imsubtract(data(:, :, 1), rgb2gray(data)). It is used to subtract red components from the grey image as shown in Fig. 7.

B = medfilt2(diff_im, [3,3]). It is used for median filtering, which is a non-linear operation often used in image processing to reduce salt-and-pepper noise. A median filter is more effective than convolution when the goal is to simultaneously reduce noise and preserve edges.

diff_im = im2bw(diff_im, 0.18). It converts the greyscale image to a binary image. The output image *diff_m* replaces all pixels in the input image with luminance greater than value one (white) and

replaces all other pixels with the value zero (black) as shown in Fig. 8.

stats = regionprops(diff_im, 'BoundingBox','Centroid'). It measures a set of properties for each connected component (object) in the binary image. The image stats are a logical array and can have any dimension.

Testing procedure

To test this program, follow the steps mentioned below.

1. Install MATLAB 7.10 (R2010a) or higher version in your system. Launch it from the desktop shortcut. You will see a blank command window.

2. Download the source code folder from EFY DVD as mentioned in EFY Note (on next page).

3. Check the device ID using command as shown in Fig. 2 and write the device ID number in the source code.

4. Run the program *detection.m*. A GUI will appear as shown in Fig. 1.

5. Click RED and the camera will start capturing the live image. The red-colour object will be detected, which is

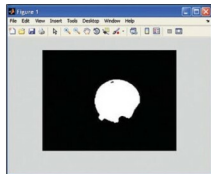


Fig. 8: Grey image to binary image

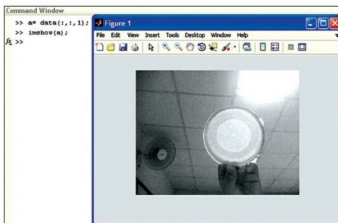


Fig. 6: Red component extracted from the original image

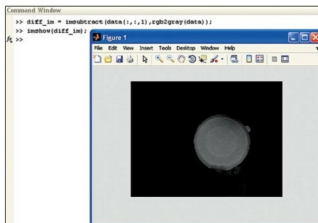


Fig. 7: Subtracting red from grey image